A Drosophila connectome substrate inside an LLM agent: what it gates, what it does not, and what it costs

Draft v1 - 2026-10-15. Applied case study with full cost accounting and negative results. Numbers marked [195] sit on a second host and must be re-derived from their artifacts before submission.

Abstract

We report an applied case study that wires a whole-organism connectome substrate (MaleCNS v1.0: 169,700 neurons, 25,582,938 innate synapses) into an LLM agent loop as a fast, cheap front layer - a digital cerebellum that acts on tasks it has consolidated and escalates to the LLM otherwise. Two experiment lines are reported. In a foraging simulation, closed-loop evolution lifted population survival from 13/24 at generation 0 to 24/24 at generation 9 (single lineage), with mean energy 6.8x higher, and the evolved population transferred to an unseen world seed at 24/24 with higher energy still. In an applied task line (8-fly populations, an 18-task operational battery plus 3 held-out tasks, three arms totalling 1,029 generations, 26,678 LLM calls, 17,005,656 tokens), the substrate's one measurable contribution is a gate: under a contrast readout the gate is alive from generation 4 and sustains 40-55 of 144 possible solo (zero-token) task executions per population per generation, with a best fly gating 8 of 18 tasks. All 24 held-out solves came from fresh LLM code generation (320-334 tokens each), and a fly with 365 learned synapses on an essentially empty graph passed 180/180 checks: the solutions come from the LLM plus a verification loop, not from the connectome. We further report the token bill as structural (a fixed 72 calls and ~47k tokens per generation caused by cache eviction, flat across 365 generations), and a per-fly policy governor that cut LLM calls 58 to 24 (-59%) at zero meal loss in the foraging economy [195]. Three candidate roles for the substrate are judged dead by measurement: recursive memory horizon 1-2 steps, input relay equivalent to a linear copy, and zero channel margin under superposed cues. The paper is offered as an honest accounting case study: where a biological substrate helps an LLM agent, where it does not, and what the whole arrangement costs.

1. Introduction

LLM agents pay per token and pay again on every repeat of a solved problem. A natural cost cut is a front layer that decides, per task, whether the agent can act from its own consolidated competence or must consult the model. This paper asks a specific version of that question: can a biological connectome - the complete male Drosophila central nervous system graph, used as a fixed dynamical substrate - serve as that front layer, and what does the arrangement actually contribute once the accounting is done?

Contributions: (i) a working two-layer agent in which a connectome substrate sits in front of an LLM and a verification loop; (ii) a measured, bounded contribution - a gate on asking the LLM - with its denominators named; (iii) a complete token bill for 1,029 population generations; (iv) a set of falsified candidate roles stated with their numbers; (v) a per-fly policy governor that behaves as a cost container rather than a capability amplifier [195]. We believe this is the first connectome substrate run inside an LLM agent loop with the full cost ledger published.

2. Related work

Connectome simulation. Steering a virtual Drosophila descending-neuron connectome produced directed locomotion (Andlman et al., eLife 2022); the OpenWorm project simulated C. elegans neuromechanics; FlyCircuit and the FlyWire/Janelia resources supply the graphs; Shiu et al. showed that connectome networks with random weights can learn sensory-motor mapping as efficiently as weight-modified networks. Our substrate is the Janelia male connectome (MaleCNS v1.0) used as a fixed weighted graph with local three-factor plasticity on top.

LLM agent cost control. Prompt/context caching, model cascades and routers, and skill libraries (e.g. Voyager-style code skill stores) all reduce repeat cost by routing or reuse. Our gate is the same decision - ask the model or not - implemented as a spiking recurrent network over a biological graph, with the LLM writing executable skills that a verifier accepts before they count.

3. System

Substrate. N = 169,700 neurons, 25,582,938 innate weighted synapses (0.092% density), float32 throughout. Dynamics per episode, 30 steps: v = 0.9*v + 5.0*(Win@S + G@s) + drive; s = (v > 1.0); v *= (1 - s); activations accumulate. Win is the fixed innate connectome; G is the per-fly learned synapse set, updated by a local three-factor Hebbian rule (no backpropagation): potentiation and depression driven by action outcome, with sprouting and pruning under a structural cap (SYN_CAP).

Two-layer agent. Layer 1 is a per-fly skill store: for a new task the LLM writes a PoC script, a verifier runs it (exit 0, stdout length, keyword), and only verified entries are stored. Layer 2 is a 10-parameter genome per fly (plasticity and growth knobs) that is heritable; synapses and stores die with the fly. Routing per task, first match wins: solo (fly rate >= 0.01 AND verified store entry, 0 tokens); practice (verified entry, rate below threshold, 0 tokens); cap (fly LLM budget exhausted); LLM (write code, verify, store; pays tokens). A curriculum frontier advances when >= 5 of 8 flies are solo on the newest task. Evolution: 8 flies per population, rank by (solo, rate_sum, -tokens), keep 4 elites, refill with multiplicative-Gaussian mutants plus one immigrant.

Battery. 10 system-health scripting tasks plus an 8-task security queue that advances the curriculum; 3 held-out tasks (never in training) probe transfer. Tasks execute against real hosts; verification runs the produced script.

4. Experiment A: foraging simulation (economy layer)

A 24-fly population forages in an energy economy (food episodes, meals, starvation death). Closed-loop evolution lifted survival from 13/24 (gen 0) to 24/24 (gen 9), with mean energy 6.8x baseline; the plateau held to gen 19. The gen-9 checkpoint shipped with this draft verifies 24/24 flies alive. Transfer: the gen-9 population in an unseen world seed scored 24/24 with higher mean energy than at home. Selection-vs-random controls exist for five configurations (6 generations, 6-8 flies each; Fig 3): on the economy timescale the energy gap between selection and random is small and noisy at 6 generations - the survival signal (13/24 vs extinction patterns) is the carried result, not the 6-gen energy curve.

Policy comparison [195]. Four per-fly policies (baseline no-escalation; fly-first with escalation; LLM-first; governor = 50 warmup rounds then commit to fly-first or LLM-first on observed solo rate) were compared in the same economy. The governor cut LLM calls from 58 to 24 (-59%) at zero meal loss relative to baseline; it behaved as a cost container - it stopped paying for help the fly did not need - and not as a benefit amplifier. Gate-open vs gate-closed economy comparisons report meals +47% and energy +42% [195]; a teacher-transfer variant cut per-episode teaching tokens 45 to 18 [195]. All three [195] figures must be re-derived from their artifacts before submission.

5. Experiment B: applied task line (three arms)

Three populations ran to completion on the applied battery (Table 1). The headline: under a contrast readout (task rate = channel mean minus rest-of-motor mean), the gate is genuinely alive - gpu_314 breaks out at generation 4 and sustains 40-55 solo of 144 possible per generation, best fly 8 of 18 tasks free of LLM cost. gpu_313 (original genome bounds) breaks out only at generation 217, showing the bound box delayed but did not prevent the same behaviour. gpu_300 (no contrast) reads 18/18 solo per fly, but that is a gate-failure artifact: its rate_sum is +1.4178 against a 0.01 threshold, every fly identical, all synapses pinned at the structural cap - the gate is open because the meter is saturated, not because the fly can act.

Held-out probe: 8 flies x 3 never-trained tasks = 24/24 solved, every solve by a fresh LLM generation at 320-334 tokens. Harvest slot gpu_314/4 carries 365 learned synapses - an essentially empty graph - and passes 180/180 task checks: solutions come from LLM plus verifier. The solo count that does mean something is the contrast-gated one: 8/18 per fly, 40-55/144 per population.

Table 1. Three-arm harvest ledger (final state).

arm	gens	1st solo	peak solo	best fly	LLM calls	tokens
gpu_300 ungated	664	gen 0	144 (gen 506)	18/18*	78	49,554
gpu_313 orig bounds	224	gen 217	36 (gen 223)	8/18	16,461	10,429,312
gpu_314 widened	141	gen 4	55 (gen 63)	8/18	10,139	6,526,790

** gpu_300 18/18 is the gate-failure control (saturated meter), not a capability reading.*

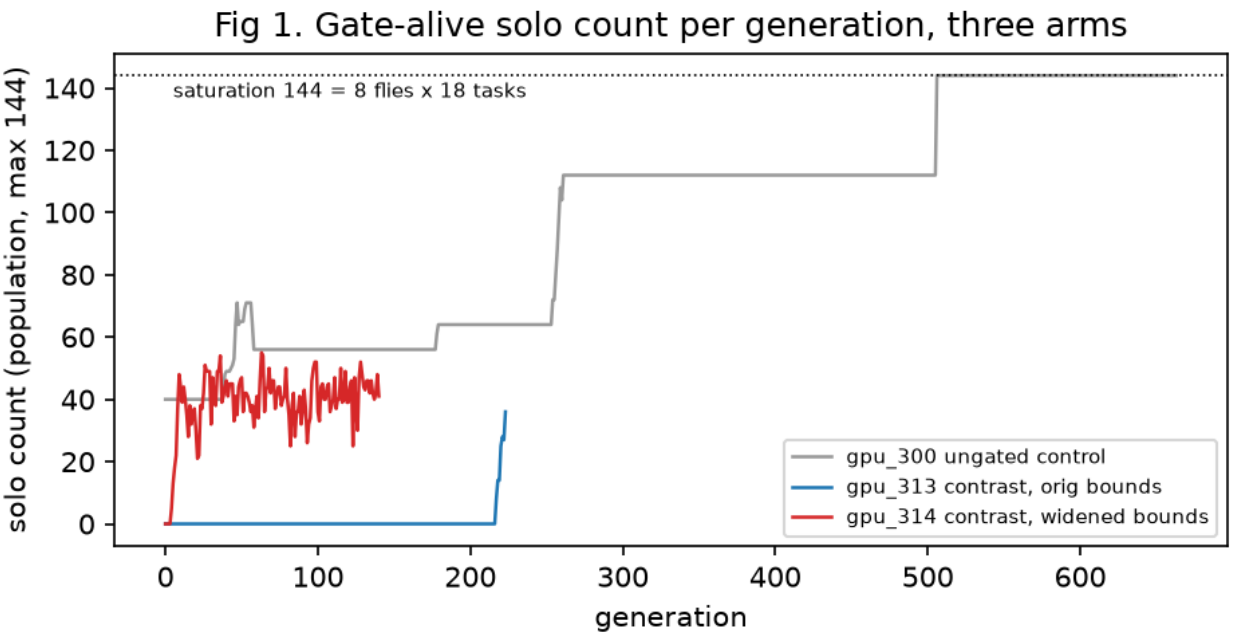


Fig 1. Population solo count per generation. gpu_300 saturates (meter artifact); gpu_314 is alive from gen 4; gpu_313 breaks out at gen 217.

6. Token economics

Totals across the two contrast arms: 26,678 LLM calls, 17,005,656 tokens. The bill is structural: from generation 0 to the end, both arms spent a flat 72 calls and ~47k tokens per generation (Fig 2). 72 = 4 emptied slots x 18 tasks: the selection scheme deletes the non-survivors' stores, so four flies rebuild their cache from zero every generation. Survivors spend 0. The LLM bill is therefore a property of our memory-eviction design, not of task difficulty - the single clearest cost lesson of the project.

The governor result [195] belongs in the same ledger: -59% calls at zero meal loss is a cost container. A structural lock-in control belongs too: gpu_300 pinned at SYN_CAP 2.5M stayed at setuptools_inject 0/8 for 165 generations; raising the cap to 3.2M moved the curriculum frontier 4 to 5 within ten minutes on the same checkpoint - capacity, not selection, was the binding constraint at that moment.

7. Negative results (the core chapter)

Three candidate roles for the substrate were judged dead by measurement, and one hypothesis was falsified outright (Table 2). We state them as results, not apologies: a front layer that cannot remember beyond 1-2 steps, relays input as a near-linear copy, and cannot separate superposed cues cannot be the decision-maker; the gate role that survives does so because it reads a coarse signal at a low rate.

Table 2. Measured negative results.

claim tested	measurement	verdict
recursive memory horizon	29 configurations, best 1-2 steps	dead
input relay as feature extractor	R^2 0.91-0.96 vs linear copy	dead
channel choice under superposed cues	margin 0.0000 (bistable: fire-all or silence)	dead
decision layer reads task identity	18 tasks x 4 amplitudes, 72 samples, 18 classes, chance 0.056: raw input 1.000, cross-task cosine 0.0017	falsified
learned readout under native gain	cross-task cosine 0.9723 vs innate 0.0618	learned, not innate
ok metric as capability	180/180 saturated incl. 365-synapse fly	no information
fly gate in ungated arm	rate -0.0007 vs threshold 0.01	never fires

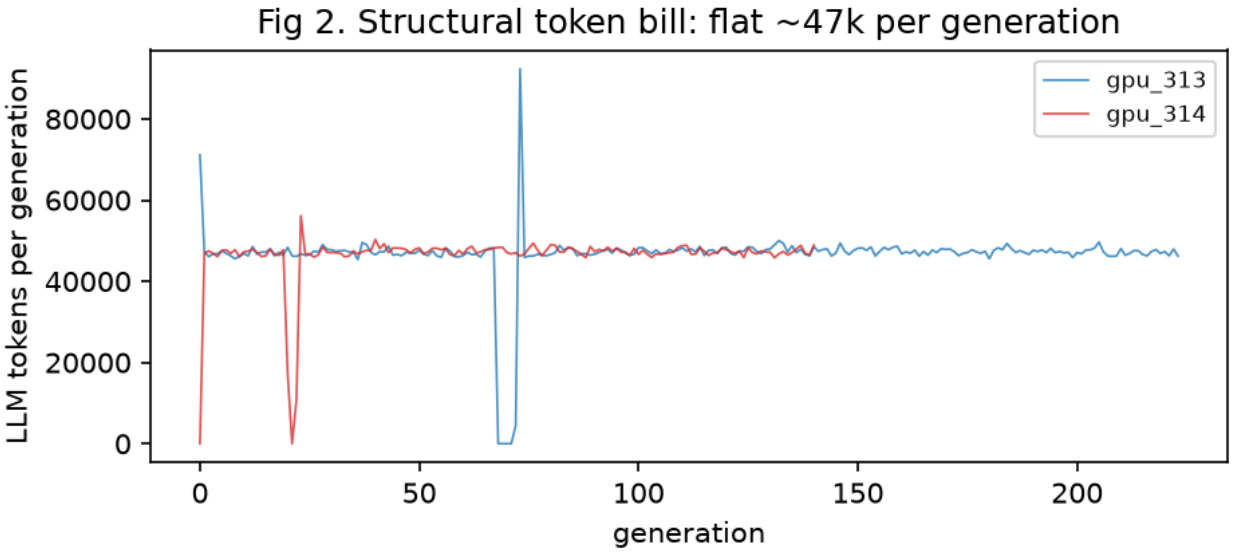


Fig 2. Tokens per generation, both contrast arms: flat ~47k - the bill is the eviction design.

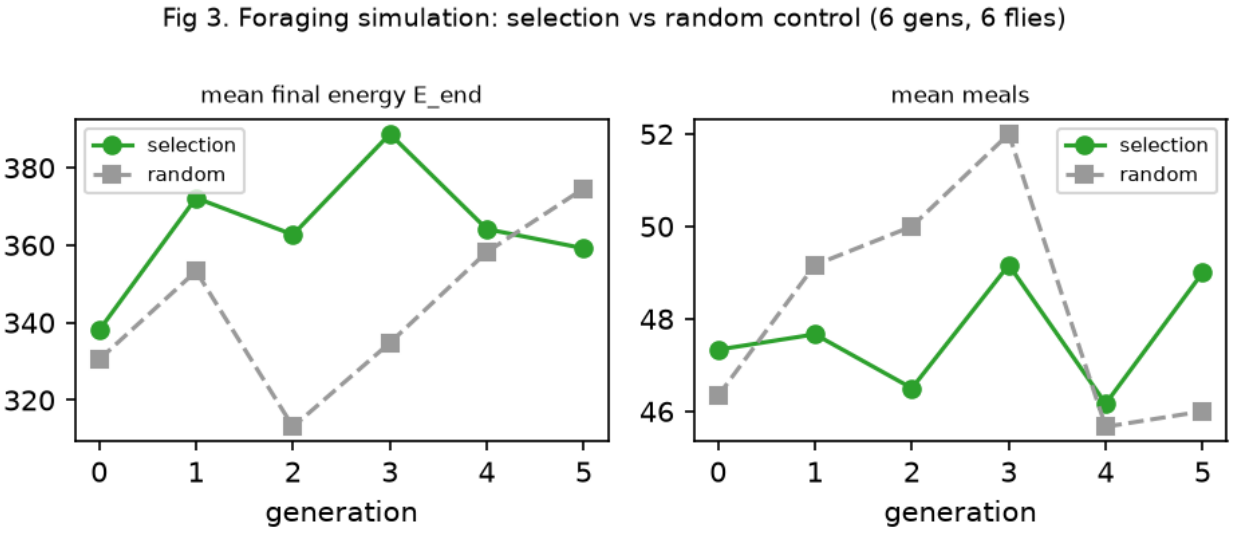


Fig 3. Foraging simulation, selection vs random over 6 generations: energy and meals tracks.

8. Discussion

Where the substrate helps is one place, and it is a real place: a cheap per-task gate signal under a contrast readout, sustaining 40-55/144 zero-token executions per population-generation and 8/18 on a single fly, with the genome evolving to keep that signal alive (breakout at gen 4 with widened bounds vs gen 217 with the original box). That is the biological analogue of a router in a model cascade - and our measurements say the router signal must be contrastive; the population-wide motor rate is a saturated meter, not a decision.

Why the decision-layer hypothesis died is mechanistically clear: 1-2 step state retention and near-linear input relay leave no substrate for per-task identity beyond what the input already carries; the classifier result (raw input 1.000 at chance 0.056) says the task label is in the cue, not in the network's transformation of it. The learned-versus-innate cosine pair (0.9723 vs 0.0618) localises the contribution to plasticity, not to the innate graph.

Limitations. The economy-lineage result is n=1 (one evolved lineage, one transfer seed). The three flagged [195] figures are quoted from artifacts on a second host pending re-derivation. Solo counts depend on store coverage; the 8/18 fly with 5,987 synapses and the 8/18 fly with 738,868 synapses show synapse count is not the driver. CUDA and MPS numerical paths differ by ~+0.06 in rate_sum at migration boundaries; cross-machine replications must pin the device. The ok metric is saturated and excluded from claims.

9. Conclusion

A whole connectome in front of an LLM agent buys one measurable thing - a gate on asking the model - and costs 17 million tokens, most of it spent on our own memory-eviction design. We publish the gate number with its denominators, the bill with its mechanism, and the dead roles with their statistics. Next: >= 3 seeds on the economy lineage, re-derivation of the [195] artifacts, and a store-retention arm predicted to drive the per-generation bill toward zero while the gate climbs.

Release: code, ledgers (evo_ledger.jsonl per arm), checkpoints and all analysis scripts will be published with the paper.